

HYPERTENSION - CLASSIFICATION AND INITIAL MANAGEMENT

Hypertension is defined as sustained systolic BP 130 mmHg or diastolic BP 80 mmHg per the 2017 ACC/AHA guidelines. Normal BP is below 120/80 mmHg. Elevated BP is 120-129 with diastolic below 80 mmHg. Stage 1 HTN is SBP 130-139 or DBP 80-89 mmHg. Stage 2 HTN is SBP 140 or DBP 90 mmHg or higher. Hypertensive crisis is SBP above 180 or DBP above 120.

Hypertensive urgency is severely elevated BP above 180/120 without evidence of acute end-organ damage. Hypertensive emergency involves severely elevated BP with acute end-organ damage including encephalopathy, acute kidney injury, MI, unstable angina, aortic dissection, pulmonary edema, or eclampsia. In urgency, reduce BP gradually over 24-48 hours with oral agents. In emergency, reduce MAP by no more than 20-25 percent in the first hour using IV agents such as labetalol, nicardipine, or hydralazine in pregnancy.

First-line antihypertensive agents include thiazide diuretics (chlorthalidone preferred), long-acting dihydropyridine CCBs (amlodipine), ACE inhibitors, and ARBs. ACE inhibitors and ARBs are preferred in CKD, diabetic nephropathy, or proteinuria. Beta-blockers are preferred in patients with prior MI, HFrEF, or AFib needing rate control. Avoid ACE inhibitors and ARBs together (dual RAAS blockade increases risk of AKI and hyperkalemia). Avoid ACE inhibitors in pregnancy due to teratogenicity (oligohydramnios, renal agenesis).

HYPERTENSION - SECONDARY CAUSES AND RESISTANT HTN

Secondary hypertension should be suspected in patients under 30 with no family history, with HTN resistant to 3 agents at maximum doses, or with sudden worsening in a previously controlled patient. Most common secondary cause overall is renovascular hypertension from renal artery stenosis. Most common endocrine cause is primary hyperaldosteronism (Conn syndrome). Screen with aldosterone-to-renin ratio (ARR). Confirm with salt loading test or fludrocortisone suppression test.

Pheochromocytoma presents with episodic headache, diaphoresis, and palpitations. Diagnosis is with 24-hour urine fractionated metanephrines (most sensitive test). Localization uses MRI abdomen. Management requires preoperative alpha-blockade (phenoxybenzamine) for at least 14 days, then add beta-blockade only AFTER alpha-blockade is established. Never beta-block first - unopposed alpha stimulation causes severe hypertensive crisis. Surgical resection is curative.

Resistant hypertension is BP uncontrolled on 3 antihypertensives including a diuretic at maximum tolerated doses. Evaluate for white coat effect, non-adherence, drug interactions (NSAIDs, oral contraceptives, decongestants), and secondary causes. The fourth-line agent of choice is spironolactone (demonstrated superiority in PATHWAY-2 trial). Renal denervation is emerging but not yet standard. Always check for sleep apnea, which contributes to resistant HTN through sympathetic activation.

ACUTE CORONARY SYNDROME - STEMI RECOGNITION AND REPERFUSION

STEMI is defined by ST elevation of 1 mm or more in 2 or more contiguous limb leads, or 2 mm or more in 2 or more contiguous precordial leads, or new LBBB with typical ischemic symptoms. Posterior STEMI shows ST depression in V1-V3 (mirror image) - confirm with posterior leads V7-V9. Right ventricular infarction complicates inferior STEMI in 30-50 percent of cases. Diagnose with right-sided leads: V4R is most sensitive (ST elevation in V4R is pathognomonic).

RV infarction pathophysiology: right ventricle depends on preload because it is thin-walled and low-pressure. Clinical triad is hypotension, clear lungs, and elevated JVP (Kussmaul sign). Management is aggressive IV fluid resuscitation. Avoid nitrates, diuretics, and morphine - all reduce preload and can cause hemodynamic collapse. Early reperfusion of the right coronary artery is the definitive treatment.

Reperfusion strategy for STEMI: primary PCI with door-to-balloon time under 90 minutes is preferred when available. Fibrinolysis is indicated when PCI cannot be performed within 120 minutes of first medical contact. Fibrin-specific agents are preferred: alteplase, reteplase, or tenecteplase. Absolute contraindications to fibrinolysis include prior hemorrhagic stroke at any time, ischemic stroke within 3 months, active internal bleeding, suspected aortic dissection, and significant head trauma within 3 months.

ACUTE CORONARY SYNDROME - POST-MI MANAGEMENT AND NSTEMI

Post-STEMI medical therapy: dual antiplatelet therapy with aspirin 81 mg lifelong plus a P2Y₁₂ inhibitor for 12 months. Ticagrelor or prasugrel are preferred over clopidogrel (superior outcomes in PLATO and TRITON-TIMI 38 trials). Prasugrel is contraindicated in prior stroke or TIA. High-intensity statin therapy (atorvastatin 80 mg or rosuvastatin 40 mg) regardless of baseline LDL. Beta-blocker: start within 24 hours if hemodynamically stable, continue indefinitely. ACE inhibitor or ARB especially if EF below 40 percent or anterior MI.

NSTEMI versus unstable angina: distinguished by troponin elevation. The HEART score stratifies risk in chest pain: History, ECG, Age, Risk factors, Troponin (each 0-2 points). Score 3 or less is low risk (0.9-1.7 percent MACE). Score 4-6 is intermediate risk. Score 7 or higher is high risk (greater than 50 percent MACE at 6 weeks). High-risk features warranting early invasive strategy (catheterization within 24 hours) include hemodynamic instability, recurrent rest angina, signs of HF, new MR, elevated troponin, EF below 40 percent.

Anticoagulation for NSTEMI: aspirin 325 mg loading dose then 81 mg daily. Add P2Y₁₂ inhibitor (ticagrelor or clopidogrel). Anticoagulate with UFH, LMWH (enoxaparin preferred), fondaparinux, or bivalirudin. GPIIb/IIIa inhibitors (eptifibatide, tirofiban) are reserved for high-risk patients proceeding to PCI not pretreated with P2Y₁₂ inhibitors. Fondaparinux is avoided as the sole anticoagulant when PCI is planned (risk of catheter thrombosis).

HEART FAILURE - CLASSIFICATION AND DIAGNOSIS

Heart failure classification by ejection fraction: HFrEF with EF below 40 percent is caused by reduced contractility from ischemia, cardiomyopathy, or myocarditis. HFmrEF has EF 40-49 percent. HFpEF with EF 50 percent or higher is caused by impaired diastolic relaxation, most commonly from HTN, obesity, and diabetes. NYHA functional classification guides prognosis and therapy decisions. BNP above 400 strongly suggests HF; BNP below 100 makes HF unlikely.

Diagnosis of heart failure: echocardiography is the definitive test for measuring EF and characterizing HF type. CXR findings in acute decompensated HF include cardiomegaly, cephalization of pulmonary vessels, Kerley B lines (interstitial edema at lung bases, perpendicular to pleura), bilateral pleural effusions (right greater than left), alveolar edema (butterfly or batwing pattern in severe pulmonary edema). The presence of Kerley B lines indicates pulmonary capillary wedge pressure above 20 mmHg.

Causes of acute decompensated heart failure are remembered with FAILURES: Forgot medications (non-adherence is the most common), Arrhythmia (new AFib), Ischemia or infarction, dietary Lapse (high sodium or fluid intake), Upregulation of renin-angiotensin (stopping diuretics), Renal failure (can precipitate volume overload), Embolism (PE), Stenosis or systemic illness (infection, thyroid disease). IV diuretics (furosemide) are first-line for acute decompensation.

HEART FAILURE - GDMT AND DEVICE THERAPY

Guideline-directed medical therapy (GDMT) for HFrEF: four drug classes with proven mortality benefit. First: ACE inhibitor or ARB - transition to sacubitril/valsartan (ARNI) in NYHA II-III patients who remain symptomatic on ACE/ARB and are hemodynamically stable. Never give ARNI within 36 hours of an ACE inhibitor (risk of angioedema). Second: beta-blocker - carvedilol, metoprolol succinate, or bisoprolol only. Atenolol and metoprolol tartrate are NOT acceptable in HFrEF.

GDMT continued: Third drug class for HFrEF is MRA - spironolactone or eplerenone. Indicated when GFR above 30 and potassium below 5.0 mEq/L. Monitor potassium closely. Fourth: SGLT2 inhibitors - dapagliflozin (DAPA-HF trial) or empagliflozin (EMPEROR-Reduced trial). SGLT2 inhibitors also benefit HFpEF (EMPEROR-Preserved and DELIVER trials) - this is the only drug class with proven benefit across the entire EF spectrum. For HFpEF, otherwise treat comorbidities (HTN, AFib, diabetes, obesity) and use diuretics for symptom management.

Device therapy in HFrEF: ICD for primary prevention when EF remains 35 percent or below after at least 3 months of optimal GDMT. The 3-month waiting period is mandatory to allow GDMT to improve EF before committing to device implantation. ICD is indicated in NYHA Class I-III but NOT in Class IV unless patient is a candidate for cardiac transplant or LVAD (bridge to transplant). CRT indications: EF 35 percent or below, LBBB morphology, QRS duration 150 ms or greater, NYHA Class II-IV despite GDMT. CRT-D (combined CRT plus defibrillator) is preferred over CRT-P for patients who also meet ICD criteria.

NYHA Functional Classification of Heart Failure

Class	Limitation	Description
I	None	No symptoms with any level of activity; normal daily function unimpaired
II	Mild	Slight limitation; comfortable at rest but symptomatic with moderate exertion (e.g., climbing stairs, brisk walking)
III	Moderate	Marked limitation; comfortable at rest but symptomatic with minimal activity (e.g., dressing, walking slowly on level ground)
IV	Severe	Symptoms at rest; any physical activity increases discomfort; unable to carry out physical activity without symptoms

ATRIAL FIBRILLATION - RATE CONTROL AND ANTICOAGULATION

Atrial fibrillation classification: paroxysmal (self-terminating within 7 days), persistent (greater than 7 days or requiring cardioversion), long-standing persistent (greater than 12 months), and permanent (decision made not to pursue rhythm control). AFib is the most common sustained arrhythmia. Key risk factors: HTN (most common modifiable risk factor), valvular heart disease (rheumatic mitral stenosis carries highest stroke risk), HF, sleep apnea, hyperthyroidism, excessive alcohol (holiday heart), and obesity.

Rate control in AFib: target resting ventricular rate below 110 bpm (lenient strategy - RACE II trial showed no benefit to strict control below 80 bpm). Preferred rate control agents: beta-blockers (metoprolol) or non-dihydropyridine CCBs (diltiazem, verapamil). Avoid diltiazem and verapamil in HFrEF - they further depress EF and worsen outcomes. Digoxin is second-line: adequate rate control at rest but insufficient during physical activity. Amiodarone can provide rate control when other agents fail but is reserved for refractory cases.

Anticoagulation in AFib: use CHA₂DS₂-VASc score. CHF (1), Hypertension (1), Age 75 or older (2), Diabetes (1), Stroke or TIA history (2), Vascular disease - prior MI or PAD (1), Age 65-74 (1), Sex category female (1). Score of 2 or higher in men or 3 or higher in women warrants anticoagulation. DOACs (apixaban, rivaroxaban, dabigatran, edoxaban) are preferred over warfarin EXCEPT in moderate-severe mitral stenosis or mechanical heart valves (warfarin only for these indications).

ATRIAL FIBRILLATION - RHYTHM CONTROL AND ABLATION

Rhythm control agents: amiodarone is most effective antiarrhythmic but carries significant organ toxicity (pulmonary fibrosis, thyroid dysfunction - both hypo and hyper, hepatotoxicity, corneal deposits, photosensitivity, peripheral neuropathy). Baseline PFTs, LFTs, and TFTs required before initiation. Dofetilide requires inpatient initiation for minimum 3 days due to QTc prolongation risk (requires continuous telemetry monitoring). Flecainide and propafenone (Class IC agents) are CONTRAINDICATED in structural heart disease including prior MI, HFrEF, and significant LVH - risk of ventricular proarrhythmia.

Dronedarone is a non-iodinated amiodarone analog with less toxicity but lower efficacy. Contraindicated in permanent AFib and decompensated HF (ANDROMEDA trial showed increased mortality). The EAST-AFNET 4 trial demonstrated that early rhythm control reduces major adverse cardiovascular events compared to rate control when initiated within 1 year of AFib diagnosis. This was a landmark trial shifting practice toward early rhythm control in eligible patients.

Cardioversion for AFib: if AFib duration is 48 hours or less, direct cardioversion is safe without TEE or prolonged anticoagulation. If AFib duration is greater than 48 hours or unknown, either perform TEE to rule out left atrial appendage thrombus before cardioversion, OR anticoagulate for at least 3 weeks before cardioversion. Continue anticoagulation for at least 4 weeks after successful cardioversion regardless of CHA₂DS₂-VASc score (due to atrial stunning and risk of thromboembolic events during recovery of atrial function).

VALVULAR HEART DISEASE - AORTIC STENOSIS

Aortic stenosis etiology by age: calcific/degenerative disease is most common in adults above 65 years. Bicuspid aortic valve is most common cause in adults 40-65 years. Rheumatic disease (always with concurrent mitral valve involvement) is most common cause in developing countries. Severe AS criteria: AVA 1.0 cm² or less, mean gradient 40 mmHg or higher, peak jet velocity 4 m/s or higher. Classic symptomatic triad: angina (3-5 year survival after onset), syncope (3 year survival), and dyspnea or HF (1-2 year survival).

AS management: no medical therapy has proven mortality benefit - beta-blockers and vasodilators are relatively contraindicated (reduce compensatory mechanisms). Definitive treatment is valve replacement. TAVR (transcatheter aortic valve replacement) is now preferred over SAVR (surgical) for high and intermediate operative risk. For low surgical risk and younger patients (below 65 years), SAVR is preferred due to longer durability of surgical valves. Bicuspid aortic valve patients generally prefer SAVR.

Indications for AVR intervention: any symptomatic severe AS (class I). Asymptomatic severe AS with EF below 50 percent (class I). Very severe AS with peak velocity 5 m/s or higher (class IIa - elective surgery before symptoms develop). Asymptomatic severe AS with rapid progression (velocity increase above 0.3 m/s per year). When in doubt in asymptomatic patients: exercise stress testing is reasonable to unmask symptoms but should be avoided in symptomatic patients (risk of hemodynamic compromise).

VALVULAR HEART DISEASE - MITRAL REGURGITATION AND OTHER VALVES

Acute severe mitral regurgitation: most common cause is papillary muscle rupture post-MI. Posterior papillary muscle is more vulnerable because it receives blood supply from a single vessel (posterior descending artery), while the anterior papillary muscle has dual supply. Presents with sudden pulmonary edema and cardiogenic shock within hours of MI. Echo shows hyperdynamic LV (EF appears artificially elevated from reduced afterload). Holosystolic murmur at apex radiating to axilla. Emergency surgical repair is required. Bridge with nitroprusside plus intra-aortic balloon pump (IABP).

Chronic MR management: surgical repair or transcatheter MV repair (MitraClip) when: EF below 60 percent (surgical threshold is 60 percent, NOT 55 percent, because MR artificially elevates EF by reducing afterload - an EF of 55 percent in severe MR represents significant systolic dysfunction), end-systolic dimension above 40 mm, new-onset AFib, or pulmonary HTN with PASP above 50 mmHg. Valve repair preferred over replacement when feasible (better outcomes, preserves subvalvular apparatus).

Mitral stenosis: most common cause worldwide is rheumatic fever. Severe MS defined by MVA 1.5 cm² or less. Classic auscultation: opening snap (OS) after S2, with short S2-OS interval indicating high LA pressure and severe stenosis. Low-pitched rumbling diastolic murmur at apex with presystolic accentuation (if in sinus rhythm). Complications: pulmonary HTN (leads to right heart failure), AFib (from LA enlargement - increases stroke risk significantly), hemoptysis, systemic embolism. Percutaneous mitral balloon commissurotomy (PMBC) preferred over surgery for pliable non-calcified valves without significant MR.

AORTIC DISSECTION - CLASSIFICATION AND DIAGNOSIS

Aortic dissection: intimal tear with blood entering the media, creating a false lumen that propagates. Stanford Type A involves the ascending aorta regardless of tear origin - surgical emergency with mortality 1-2 percent per hour if untreated. Stanford Type B involves only the descending aorta distal to the left subclavian artery - managed medically unless complicated. DeBakey classification: Type I (ascending and descending), Type II (ascending only), Type III (descending only). Stanford Type A corresponds to DeBakey Types I and II.

Classic presentation: sudden severe tearing or ripping chest pain with maximum intensity at onset (contrast with MI where pain builds gradually). Pain radiates to the back or interscapular region. BP differential greater than 20 mmHg between arms, pulse deficits (absent or unequal pulses), aortic regurgitation murmur (Type A - aortic root involvement). CXR: widened mediastinum above 8 cm in 80 percent of cases, loss of aortic knob contour, tracheal deviation to the right, pleural effusion (hemothorax from aortic leak).

Diagnostic approach: CT angiography of chest/abdomen/pelvis is the test of choice for stable patients - fast, widely available, high sensitivity. TEE is the alternative for unstable patients where CT is not feasible (can be performed at bedside or in OR). MRI has highest sensitivity but time-consuming and not practical in emergency. Transthoracic echo can detect Type A dissection but has limited sensitivity. Key risk factors: chronic hypertension (most common modifiable RF), Marfan syndrome (FBN1 mutation), bicuspid aortic valve (associated aortopathy), Turner syndrome, cocaine use, pregnancy.

AORTIC DISSECTION - MANAGEMENT AND MARFAN SYNDROME

Type A aortic dissection: emergency surgical repair is required. Without surgery, 50 percent die within 48 hours and 90 percent die within 3 months. Pre-operative medical management: IV beta-blockade (esmolol) to target HR below 60 bpm and SBP 100-120 mmHg. After HR is controlled, add vasodilators if needed (nicardipine or nitroprusside). Never give pure vasodilators without prior beta-blockade - reflex tachycardia increases aortic wall shear stress and can propagate the dissection.

Type B uncomplicated dissection: impulse control therapy with IV esmolol or labetalol targeting HR below 60 bpm and SBP 100-120 mmHg, then transition to oral agents. TEVAR (thoracic endovascular aortic repair) is indicated for complicated Type B: end-organ malperfusion (mesenteric, renal, or limb ischemia), persistent or recurrent pain despite medical therapy, rapid expansion on imaging, or impending rupture. TEVAR covers the primary intimal tear with a stent graft to promote false lumen thrombosis.

Marfan syndrome: autosomal dominant FBN1 mutation causing fibrillin-1 deficiency. Cardinal features: tall stature, arachnodactyly (long fingers), ectopia lentis (lens dislocation - upward), pectus excavatum, high arched palate, arm span greater than height, and aortic root dilation. Annual echocardiography to monitor aortic root. Prophylactic aortic root replacement when root reaches 5.0 cm (4.5 cm in women or rapidly expanding more than 5 mm per year). Beta-blockers slow aortic dilation rate. Losartan (ARB) may have additive benefit by blocking TGF-beta signaling. Avoid contact sports and isometric exercise.

HIGH-YIELD CARDIOLOGY BOARD PEARLS - SUMMARY

HTN emergency versus urgency: emergency has end-organ damage (encephalopathy, AKI, MI, aortic dissection, pulmonary edema, eclampsia) - treat with IV agents, reduce MAP by less than 25 percent in first hour. Urgency is severely elevated BP without end-organ damage - treat with oral agents, reduce gradually over 24-48 hours. Common IV agents: labetalol (alpha and beta), nicardipine (CCB), hydralazine (safe in pregnancy), phentolamine (alpha-blocker - first choice in pheochromocytoma crisis), nitroprusside (potent vasodilator - risk of cyanide toxicity with prolonged use).

RV infarction complicating inferior STEMI: diagnose with right-sided leads (V4R most sensitive). Classic triad: hypotension, clear lungs, elevated JVP. Treatment: IV fluids (preload-dependent RV). AVOID nitrates, morphine, and diuretics. Reperfuse the right coronary artery urgently. SGLT2 inhibitors benefit both HFrEF and HFpEF - the only drug class proven across the full EF spectrum. ICD requires EF 35 percent or below after at least 3 months of GDMT and NYHA Class I-III.

Flecainide and propafenone (Class IC antiarrhythmics) are CONTRAINDICATED in structural heart disease including prior MI, HFrEF, and significant LVH - risk of fatal ventricular proarrhythmia. Safe only in patients with lone AFib (no structural disease). Pheochromocytoma management: always alpha-block FIRST with phenoxybenzamine (10-14 days minimum), then add beta-blocker. Reversing this order causes unopposed alpha stimulation and hypertensive crisis. Type A dissection requires emergency surgery. Type B uncomplicated: IV beta-blockade first, then vasodilators if needed. Never give pure vasodilators first.

Figure 1: Cardiovascular Risk Stratification Categories

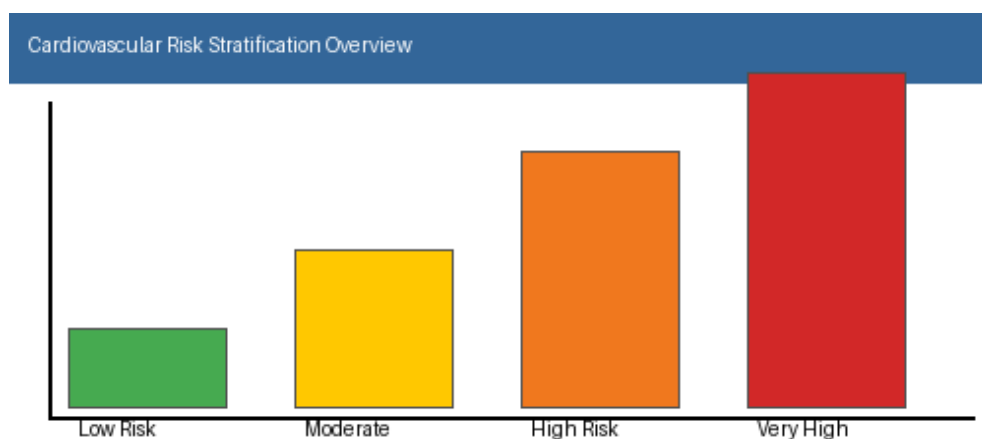


Figure 1. Risk stratification guides treatment intensity decisions in cardiovascular disease management.